CASCADA

Water Scarcity in Mexico City 2054

Group 12 - Peter Childs

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Future Scenario

In 2054, water scarcity dictates daily life. Strict usage limits are enforced, and households compete for limited supplies. Intermittent shortages persist, forcing reliance on costly water deliveries. The city’s ageing infrastructure intensifies the urgency of water-saving measures.

Contextualisation

Geographical

- Over-extraction of groundwater is leading to severe **land subsidence**, damaging infrastructure and reducing the capacity of aquifers to recharge effectively in the future.
- City's reliance on diminishing aquifers makes it particularly **vulnerable** to long-term changes in rainfall patterns.

Social

- **Access to clean water** is already a source of tension, and social unrest could escalate if the scarcity worsens in the future.
- The increasing density of **informal settlements**, many of which lack proper water infrastructure, intensifies the crisis and creates significant inequities in water access.

Political

- **Governance & Fragmentation:** Water management in Mexico City (CDMX) involves multiple agencies and overlapping jurisdictions, leading to inefficiencies.
- Water importation from surrounding basins such as the Lerma and Cutzamala systems has sparked **political tension** with other states, which view this as exploitation of their resources.

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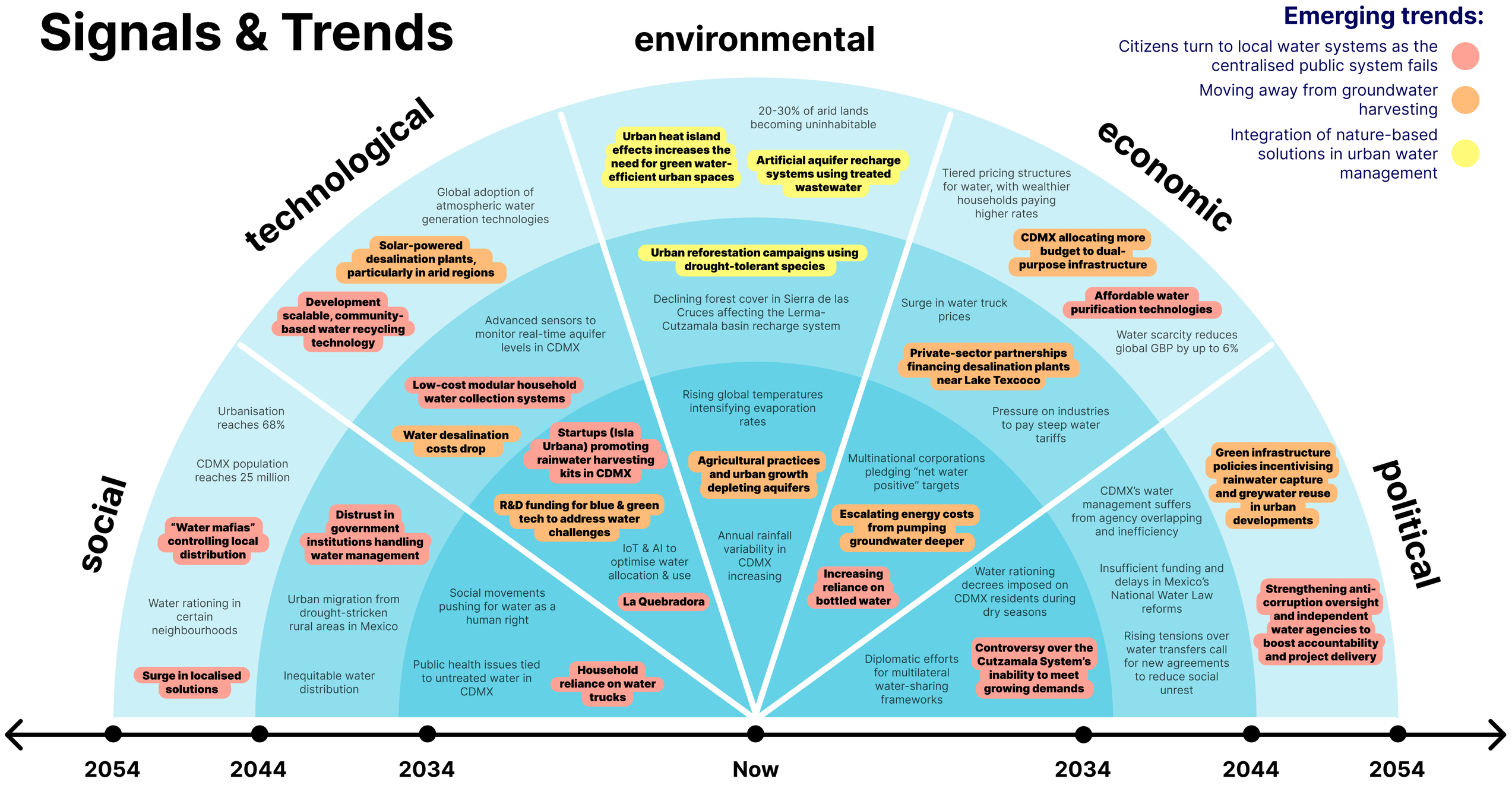
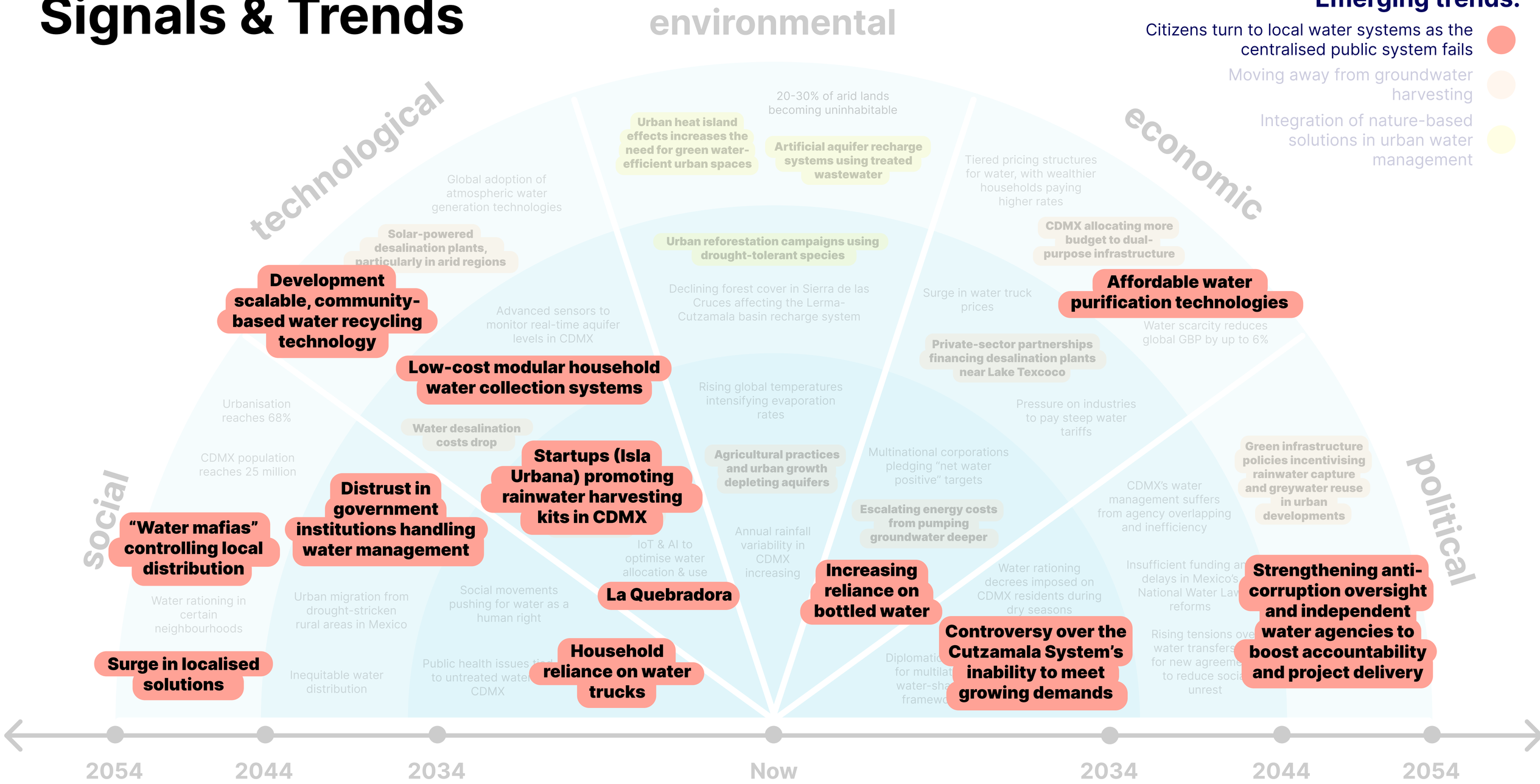
Signals & Trends



Emerging trends:

- Citizens turn to local water systems as the centralised public system fails
- Moving away from groundwater harvesting
- Integration of nature-based solutions in urban water management

Signals & Trends



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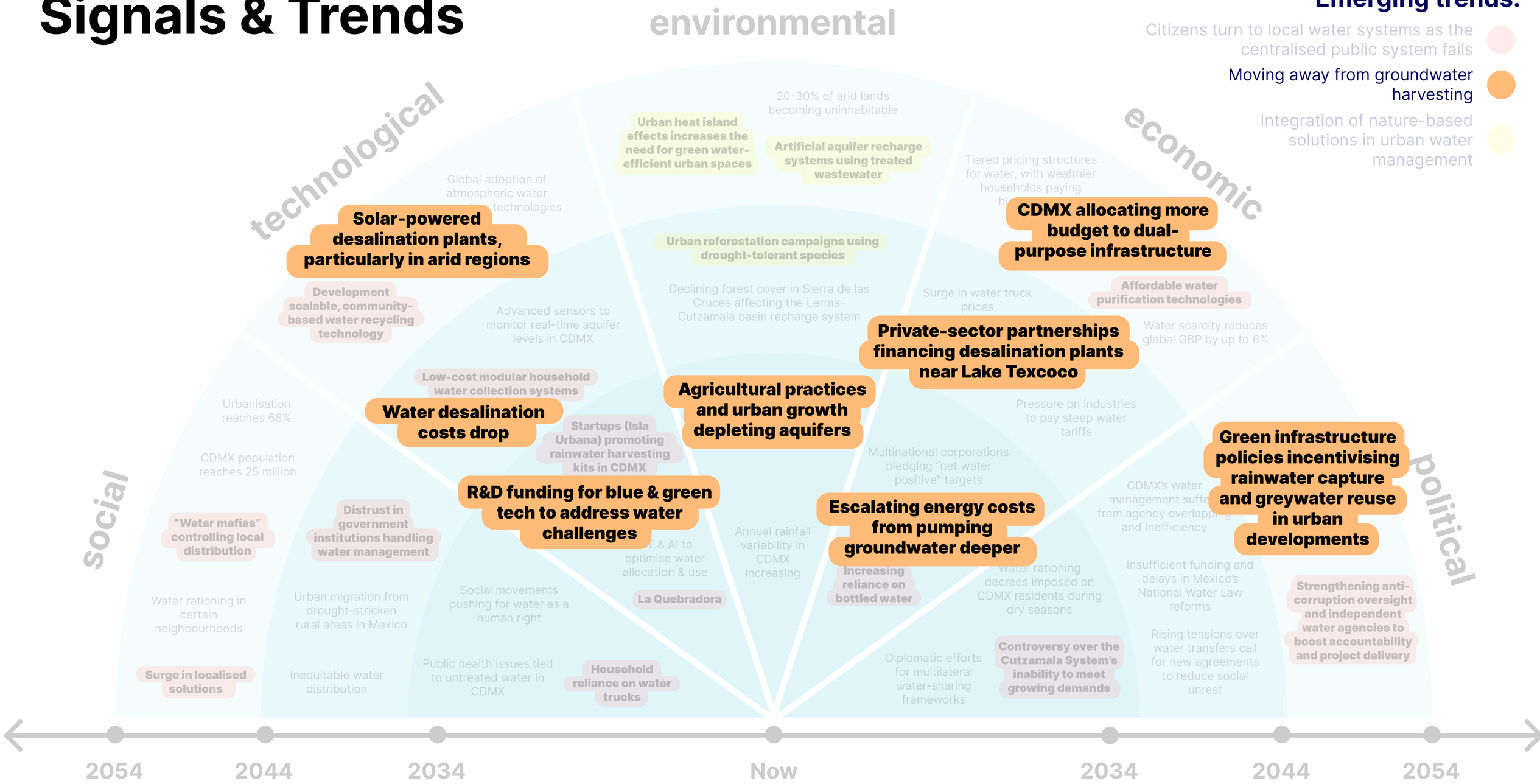

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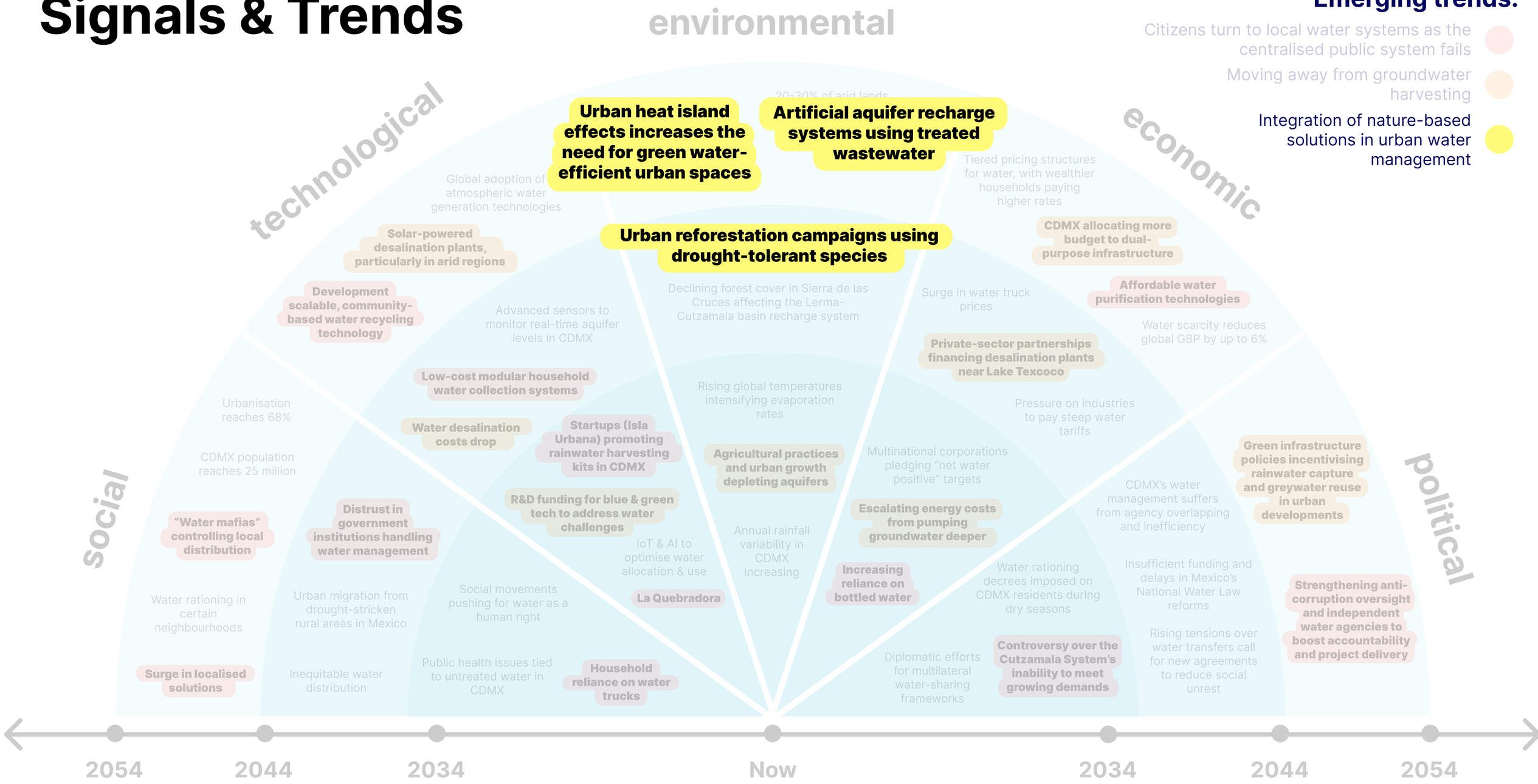

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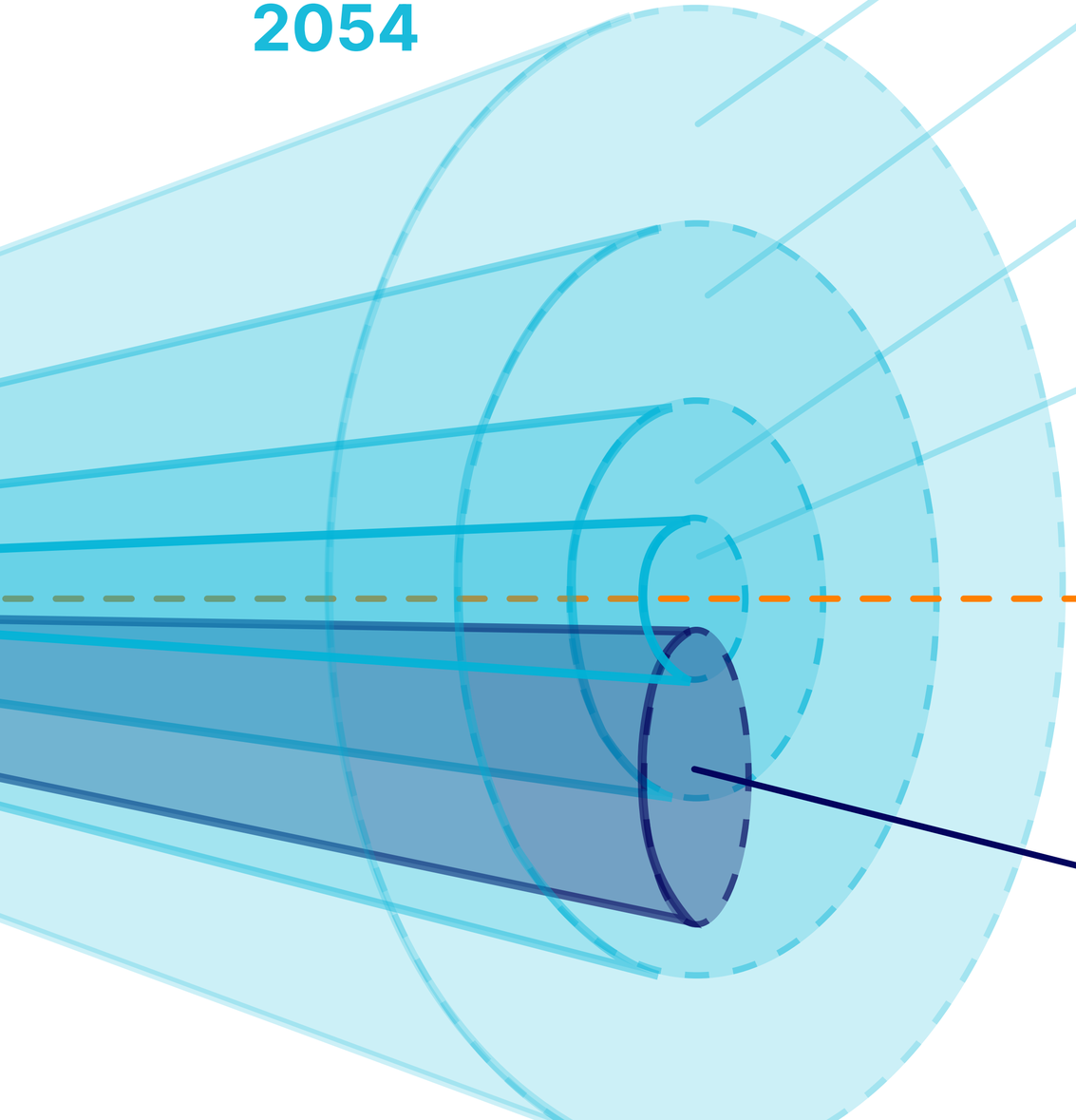
Futures Cone

2054



Preposterous

- Complete population dissolution
- Floating city
- Large scale chemical water generation

Possible

- Shift to 100% desalinated seawater supply
- Wide-scale implementation of Atmospheric Water Generation (AWG) technologies
- Excavation of old water canals and rebuilding dams to reintroduce running river water

Plausible

- Regional water-sharing agreements reduce over-dependence on aquifer
- Adoption of large-scale wastewater recycling and greywater systems

Probable

- Unequal access to water as public prices increase and private supply is expensive
- Government enforced water rationing

Projected

- Continued reliance on aquifers, worsening subsidence and infrastructure
- Moderate but insufficient investment in infrastructure

Preferred & Probable

- Adoption of hydro-sensitive urban design to improve water capture
- Decentralised rainwater capture and filtration, reducing households' dependence on public infrastructure and government plans

Future Scenario Persona

https://www.istockphoto.com/photo/iztapalapa-mexico-city-gm1432402528-474663963
https://www.bgs.ac.uk/news/understanding-the-resilience-of-mexico-city-to-future-water-crises/

Iztapalapa (Southeast)

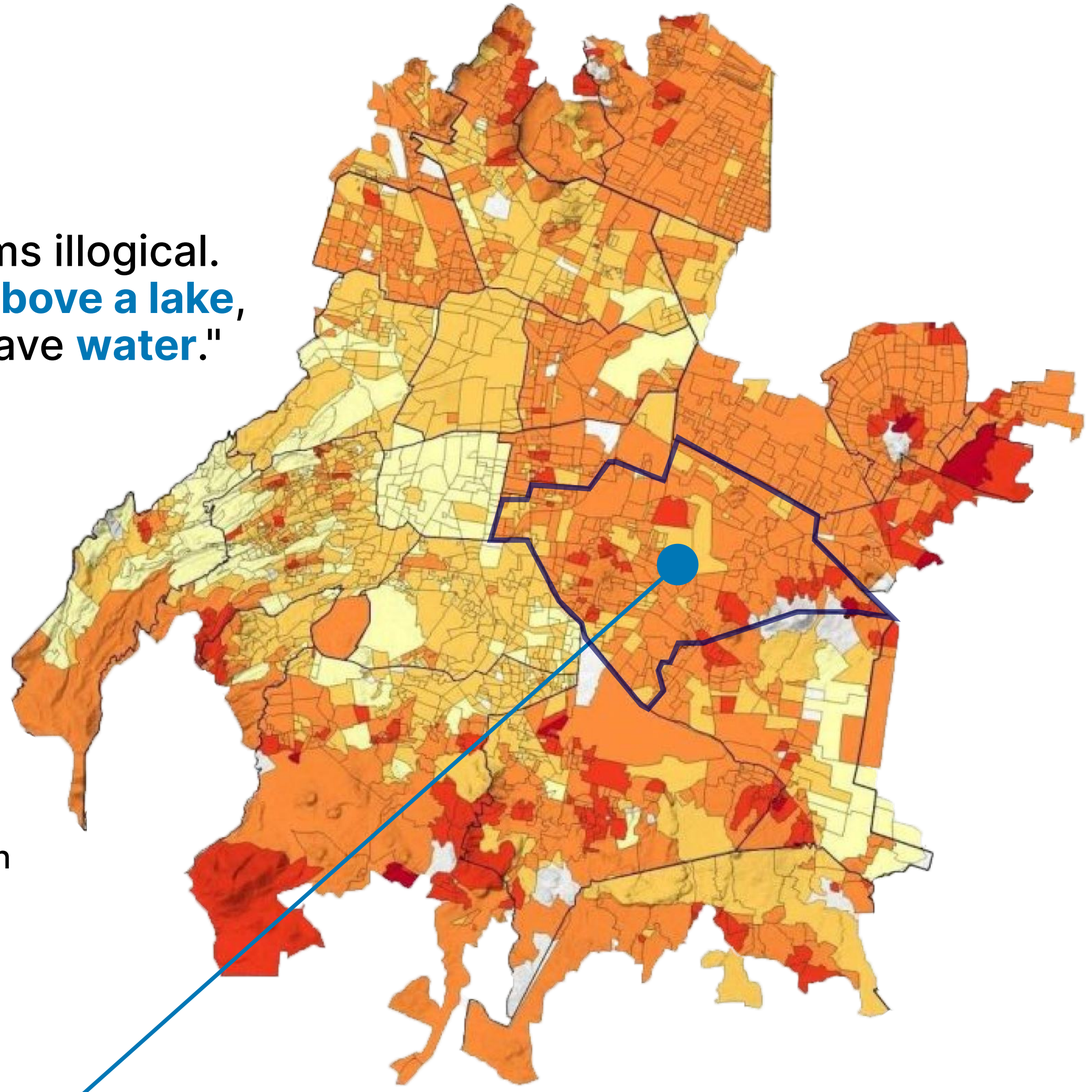
Population: 1,835,486
Total Area: 116,670,000 Sqm
Level of poverty: High
Water Truck Dependency: 40% of households
Daily Consumption per Capita: 40 litres/day

→ significantly below the Mexico City average and WHO's 100 litres/day recommendation

"To me, it just seems illogical. This city was **built above a lake**, and yet we don't have **water**."

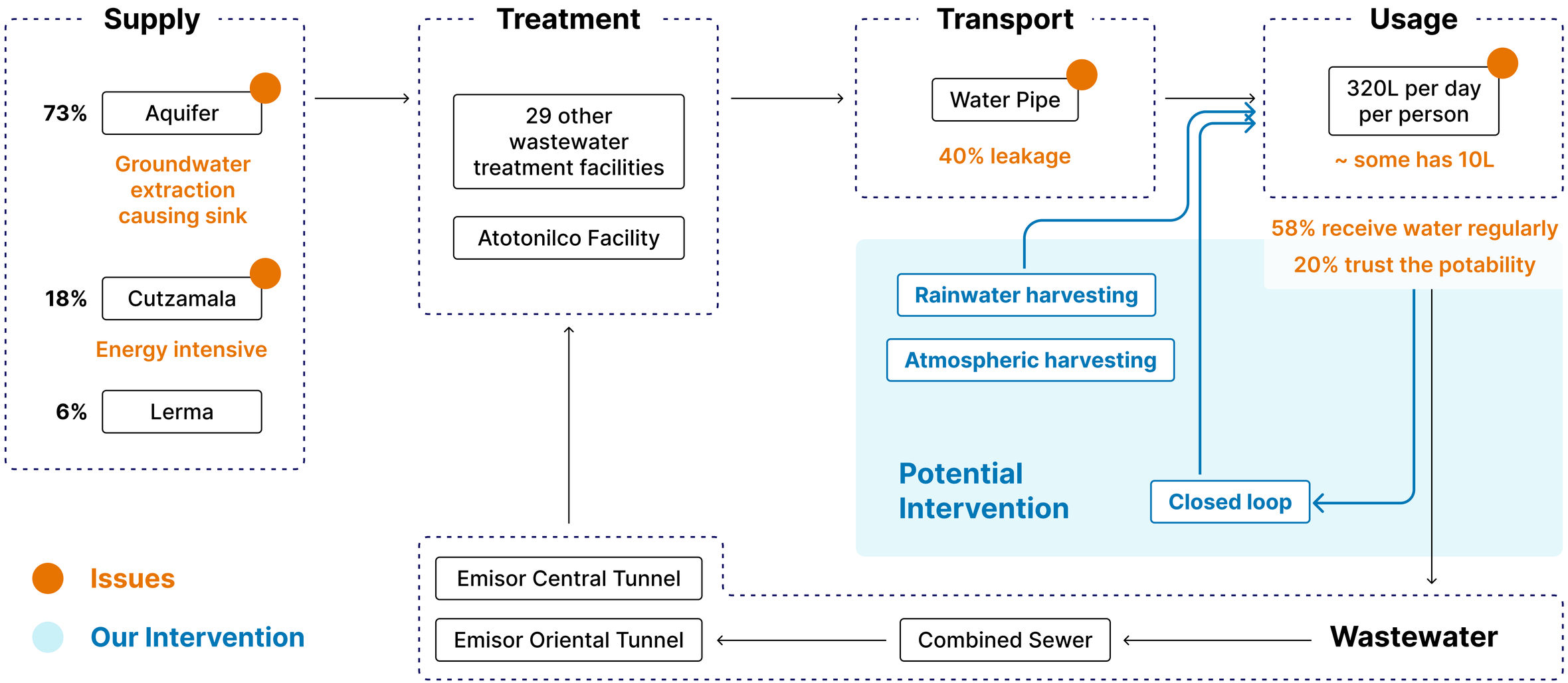
Critical Water Shortage

- High levels of socioeconomic marginalization and poverty
- Residents rely on municipal water trucks, or pay for private trucks.
- Some residents illegally tap into water lines

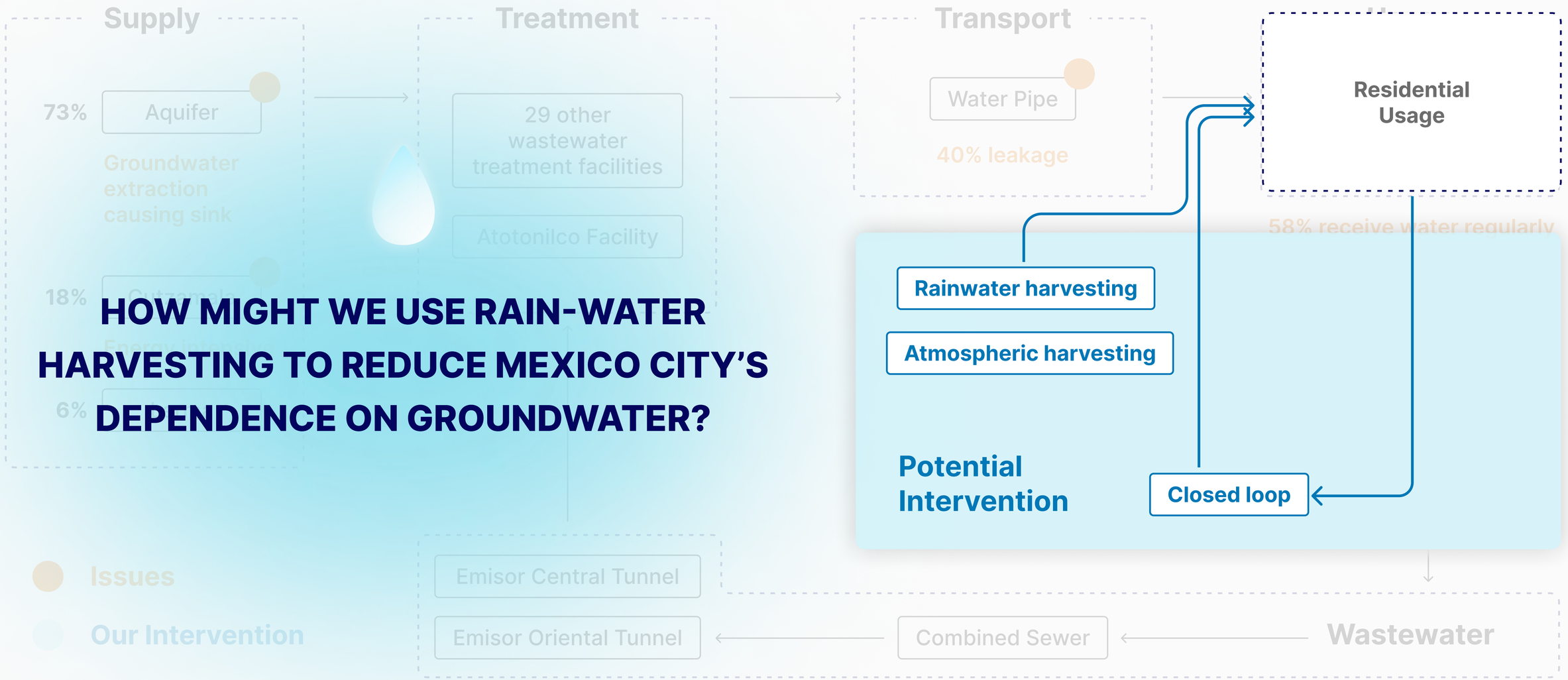


Map showing the hydrological resilience of Mexico City

Current System Diagram



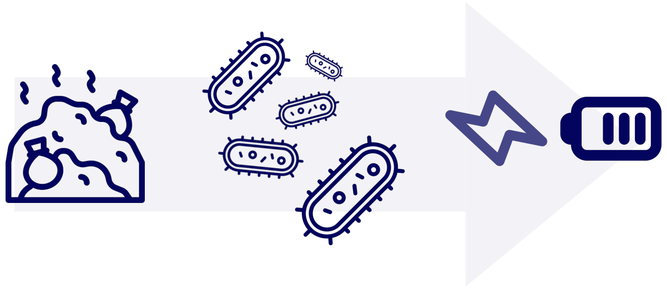
Design Engineering Opportunity



3 Key Enablers

BES (Bioelectrochemical systems)

Water purification / Energy production



- From the RCS Advances
- Using bacteria to **filter** water while producing **electricity**.
- Allows to create **decentralised system**
- Currently it's only been tested in **lab** settings

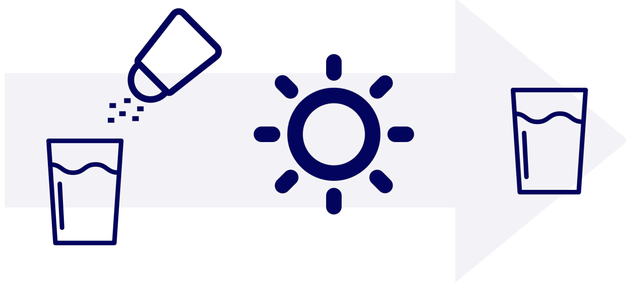
Nano-Enhanced Filtration Membranes

Water purification

- The different applications of **nanotechnology** for filters.
- **Dendrimers** deal with heavy metals like arsenic.
- **Nano-membrane filtration** removes microbes and small pollutants.

MIT Desalination system

Water cheaper than tap water

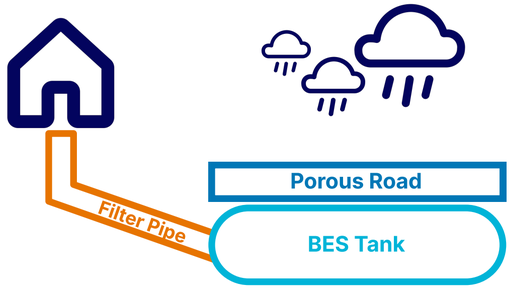


- MIT's **photovoltaic electrodialysis** desalinates water
- Very cheap, using the **sun** and **currents** to avoid clogging
- Tested in New Mexico under **real world** conditions.

3 Key Concepts

Porous pavements

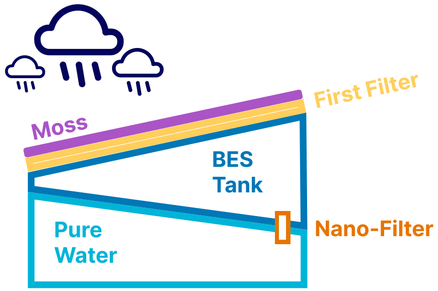
Water collection under roads



- The porous road acts as an **initial filter** into tanks below.
- These then filter using BES and nano-membranes, and **pumped** into adjacent homes as **potable** water without having to rely on the existing water infrastructure

Rooftop Moss Catchers

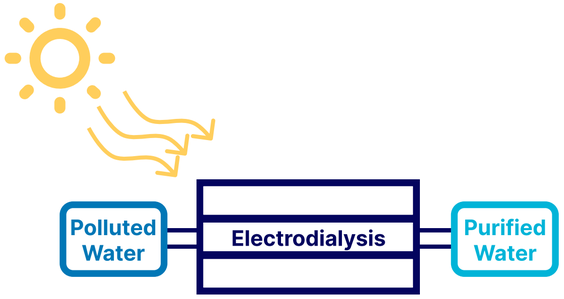
Water collection on roofs



- **Multiple layers** collect, filter, purify and filter the water again until it's potable.
- Starting with **moss**, we'd move into BES and nano-membranes for purification.
- Allows for **decentralisation**.

Home Desalination

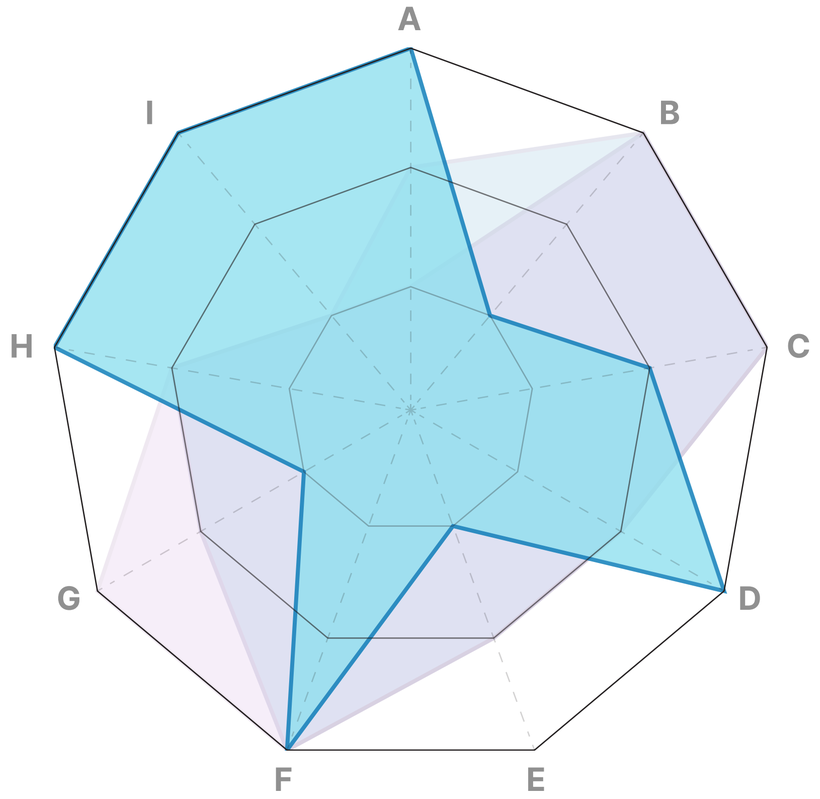
Water purification system



- Rainwater could both be filtered through **MIT's desalination** technique, removing heavy metals, debris and larger pollutants.
- **Struggles** with microbial, chemical and grease pollutants

Concept Evaluation

	Evaluation Criterias	Weight
A	**Durability**	11.11%
B	**Affordability**	11.11%
C	**Scalability**	2.78%
D	**Ease of Use** Minimal user intervention in operation	19.44%
E	**Ease of Installation**	11.11%
F	**Water Quality** Processed water meets the standard for potable water	22.22%
G	**Decentralisation** Operate independently, resilient to disruption in central system	13.89%
H	**Size**	5.56%
I	**Monitoring & Control**	2.78%



Porous Pavement	Rooftop Moss Catcher	Desalination
2.25	2.47	2.36

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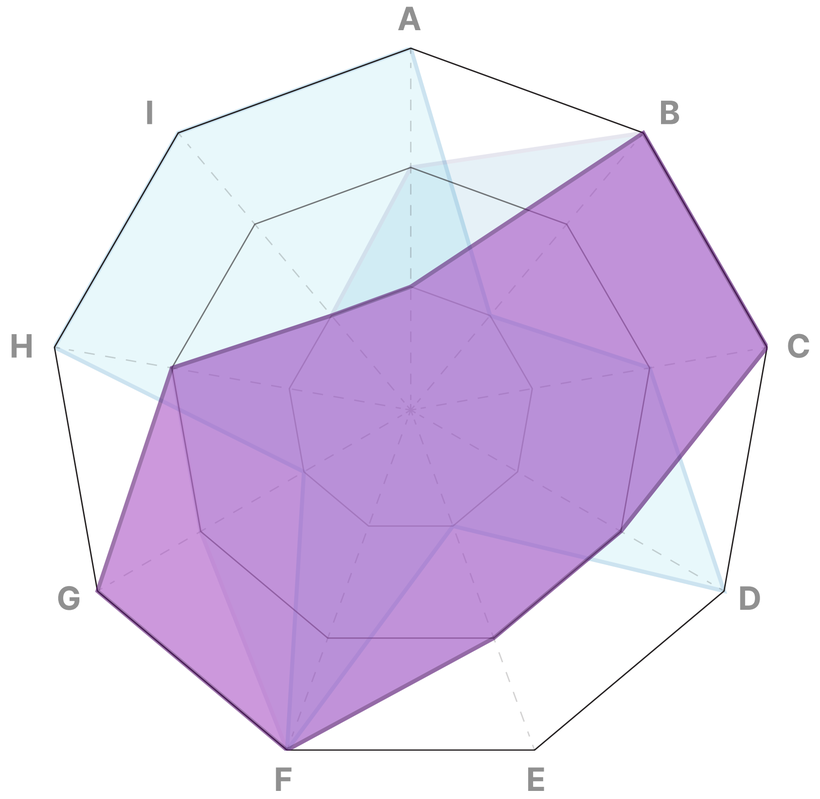

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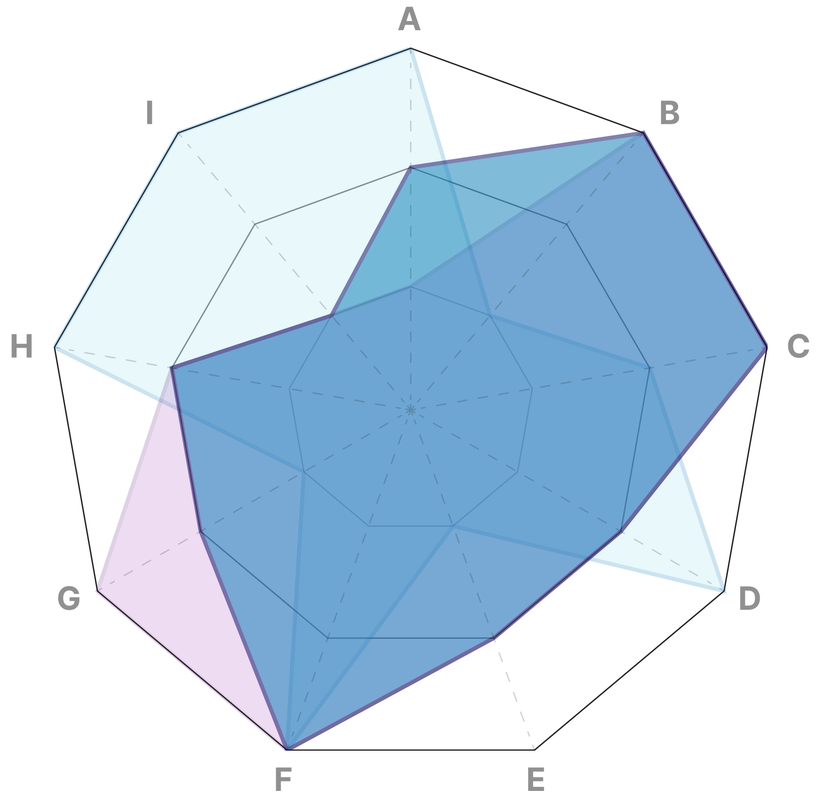

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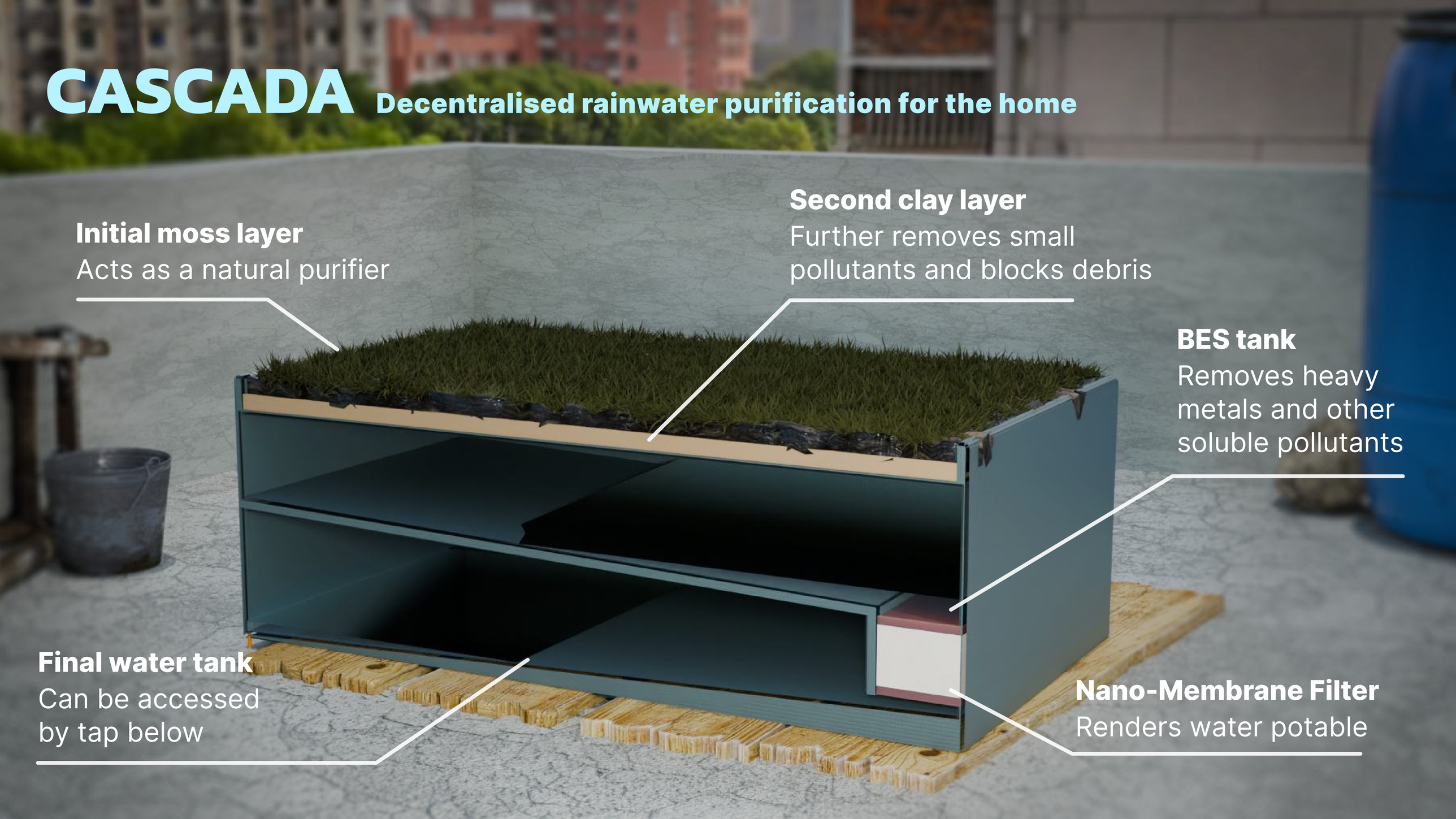
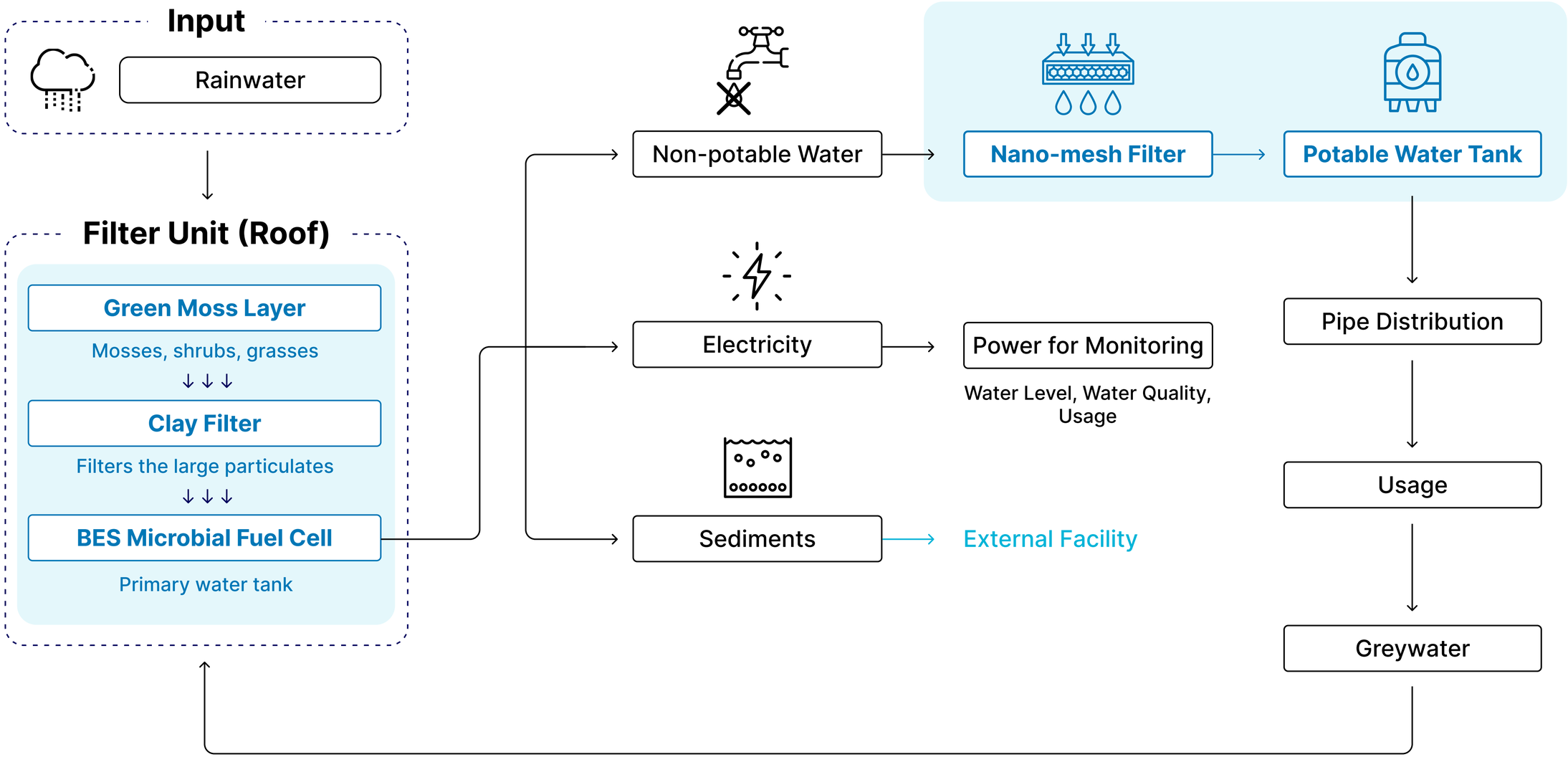
Decentralised rainwater purification for the home



System Diagram



Next Steps

Plans for Embodiment

Concept Validation

Refine based on expert feedback and feasibility study

Mid-January

Dr. Dominik Weiss

ICL Professor of Environmental Geochemistry,
Focusing on pollutant detection and sustainable water systems

Technical

Explore how BES-generated energy **powers monitoring systems**

Optimal BES selection

Identify the **most efficient BES** for pollutant breakdown and energy generation

Qilei Song

ICL Reader in Chemical Eng,
Specializing in membrane for water purification

Optimise **BES and nano-mesh filter combinations** for efficiency

Mid-February

Design

Stakeholder & Expert Consultation

Interview local agencies to understand water needs and infrastructure challenges

Develop renders to showcase functionality and gain stakeholder support

Mid-February

Incorporate feedback to improve usability and adapt the system for local conditions

Combine technical and design insights into a cohesive concept for validation

Reflection

Francesco

Key Struggles: Indecisiveness was a major hurdle as everything from picking a scenario to concept selection took very long.

Personal Development: Working with research paper to learn about new technologies. How to create grass in Blender, with improvements in vertex editing and UV mapping.

Erik

Key Struggles: Stacking fatigue from continuous meeting

Personal Development: Got a deep understanding of Mexico's history, both ancient and modern, and its relation with the water problem. Surprised to find droughts are not only an environmental issue.

Constance

Key Struggles: Selecting a single scenario from all the research conducted and finding small-scale interventions to remediate a large-scale problem

Personal Development: Developed future building skills, learning how to analyse individual articles and derive trends and potential impacts.

Yujeong

Key Struggles: Integrating key enablers and new technologies into a workable concept

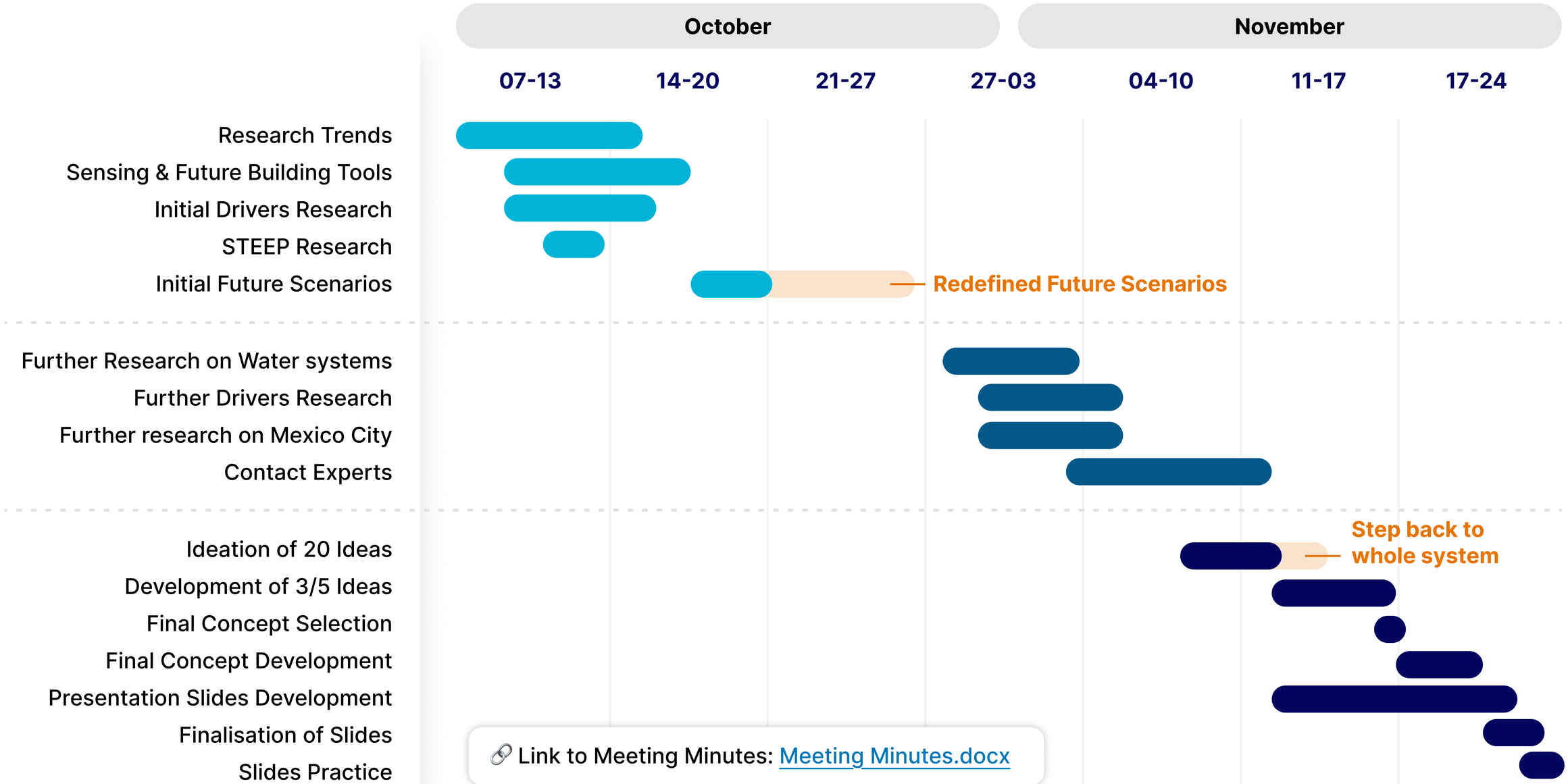
Personal Development: Developing and iterating on system diagram; ability to look into the entire system and how components are connected, rather than coming up with a standalone solution

Yang

Key Struggles: Struggled with effectively expressing ideas and lacked strong subjective opinions during the early decision-making process.

Personal Development: Improved ability to review and synthesize technical papers especially in BES, gaining a clearer understanding of workflows and technical systems.

Gantt Chart



🔗 Link to Meeting Minutes: Meeting Minutes.docx

TIME FOR QUESTIONS
https://unequalscenes.com/mexico

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Supplemental Pages

Why Mexico City?

Geographical

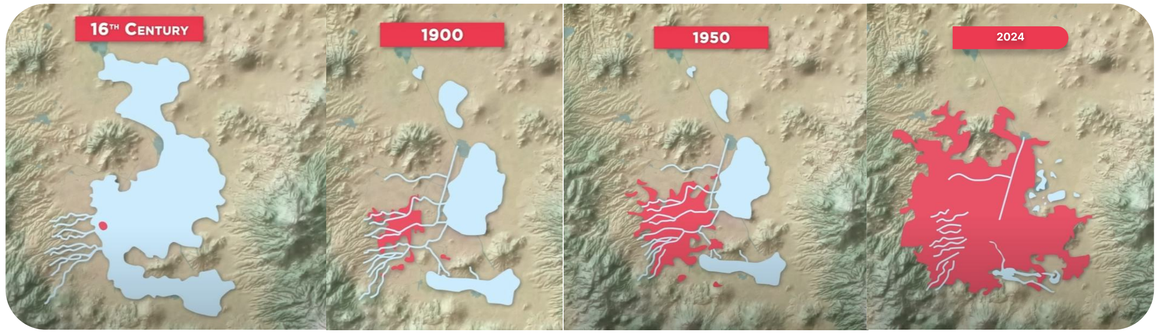
- Mexico City is located in a region where rising global temperatures are expected to intensify droughts. By 2050, climate models predict significant reductions in precipitation during the dry season, while extreme rainfall events may become more frequent, leading to challenges in water storage and flood management.
- The city's reliance on diminishing aquifers makes it particularly vulnerable to long-term changes in rainfall patterns.
- High altitude and mountainous terrain surrounding Mexico City render the importation of water from distant regions an energy-intensive endeavour.
- Over-extraction of groundwater is leading to severe land subsidence (sinking). This not only damages infrastructure but also reduces the capacity of aquifers to recharge effectively in the future.

Social

- Mexico City is projected to experience continued urban expansion, adding to the demand for water in both domestic and industrial sectors.
- The increasing density of informal settlements, many of which lack proper water infrastructure, intensifies the crisis and creates significant inequities in water access.
- Wealthier neighbourhoods often have consistent access to water, while marginalized communities rely on irregular and expensive deliveries by water trucks, deepening social divides.
- Access to clean water is already a source of tension, and social unrest could escalate if the scarcity worsens in the future.

Political

- Governance & Fragmentation: Water management in Mexico City involves multiple agencies and overlapping jurisdictions, leading to inefficiencies. Political disputes between local, state, and federal governments further hinder long-term planning and equitable distribution.
- Decades of underinvestment in water infrastructure have left the city with a crumbling network of pipes and reservoirs, with nearly 40% of water lost to leaks.
- Water importation from surrounding basins such as the Lerma and Cutzamala systems has sparked political tension with other states, which view this as exploitation of their resources.
- Efforts to implement large-scale desalination plants, rainwater harvesting, and wastewater recycling face political hurdles, including resistance from vested interests and limited funding.
- Balancing urban growth with water sustainability remains politically contentious, particularly given the need to accommodate a projected population of over 25 million by 2050.



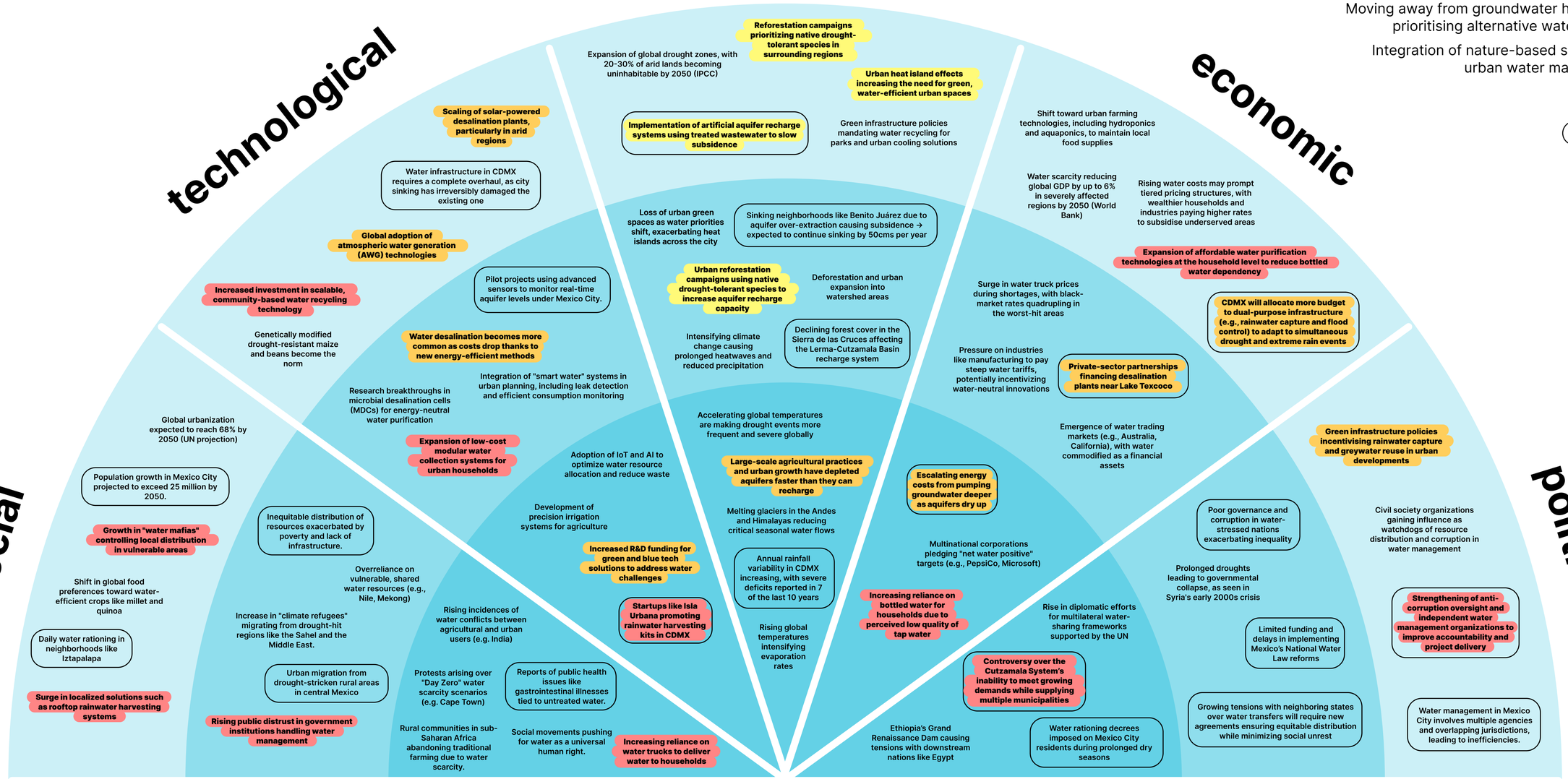
Trends & Signals

environmental

Emerging trends:

- Adoption of decentralised water systems, due to lack of trust and inefficient public infrastructure
- Moving away from groundwater harvesting, prioritising alternative water sources
- Integration of nature-based solutions in urban water management

Mexico related



Initial Scenario

DRIVERS / EVENTS / TRENDS What forces and issues are converging?	IMPACTS/IMPLICATIONS What might happen as a result?	PERSONA Who or what is the focus of this scenario?	EMERGING EXPECTATIONS What are their needs and concerns?
Droughts and floods are becoming more and more common as a result of climate change. More urban buildings and reduction of green spaces disturb the absorption of water.	Previously unaffected regions will have to suddenly overcome severe water events. Cities and residents have to recover from infrastructure damage.	Low to medium income households in areas that are affected by frequent flooding and long lasting draughts, such as the Mediterranean basin.	Water scarcity leads to increased costs for water and resources linked to it. Water over-abundance can lead to flooding which damages property, but also to poor water quality as pollutants are transported.

SCENARIO Describe a slice of the future where the forces above come together	DES ENG OPPORTUNITY How are you meeting the needs of this scenario?
The government increases its investment in flood resilient infrastructure and water regulation systems. Green cities which absorb stormwater runoff, and drought resistant measures are universalised. Populations start migrating away from the coast in fear of hydrological disasters. Water rationing can be observed in regions with serious droughts.	• A product or system to regulate the flow of the water in urban landscapes • A product which deals with specific issues, such as stormwater runoff, which contaminates nearby areas with pollutants picked up in the urban streets.

Design Statement:

We aim to design a system that effectively manages water flow in urban areas to prevent flooding and dealing with excess water, possibly for use during droughts.

Potential ENABLERS What tech is helping you to achieve this?
• Sponge Cities in China which absorb excess water. • Rice terraces. • Low cost concrete reservoirs to store water. • AI technology integrated into urban sewer system (University of Sheffield) • Floating architecture

Refined Scenario

DRIVERS / EVENTS / TRENDS What forces and issues are converging?	IMPACTS/IMPLICATIONS What might happen as a result?	PERSONA Who or what is the focus of this scenario?	EMERGING EXPECTATIONS What are their needs and concerns?
Mexico City's growing population, subsidence from over-extracted aquifers, and unpredictable rainfall amplify water scarcity. 40% of water is lost via pipe leakage and 48.4% of water sources are classed as highly polluted. Alternative infrastructures are insufficient to deal with the crisis.	Land subsidence damages water infrastructure, making it more difficult to deliver water reliably. Repairing damage and sourcing alternative water supplies increases the cost of water, impacting affordability for residents.	Low- to medium-income households in Mexico City, vulnerable to water scarcity, price increases, and limited access to clean water.	Residents expect integrated stable water systems that conserve resources, reduce bills, and operate independently, while fitting seamlessly into high-density building infrastructure.

SCENARIO Describe a slice of the future where the forces above come together	DES ENG OPPORTUNITY How are you meeting the needs of this scenario?
In 2054, water scarcity dictates daily life. Strict usage limits are enforced, and households compete for limited supplies. Smart taps, low-flow showers, and greywater systems manage daily needs, but intermittent shortages persist, forcing reliance on costly water deliveries. The city's ageing infrastructure intensifies the urgency of water-saving measures.	Develop a resilient, decentralized water management system capable of handling water flow efficiently include: • Being resilient against external damage factors • Ensuring internal water saving and recycling

Design Statement:

We aim to design a system that can deal with the water flow issues around Mexico city in order to render its low to medium income households less vulnerable to the water crisis and its increased costs.

Potential ENABLERS What tech is helping you to achieve this?
Earthquake resistant pipe by Kubota Earthquake proof bridges in Mexico (source) Reduce water loss using advanced pressure management (source) Use of nanomaterials for the removal of pollutants from wastewater

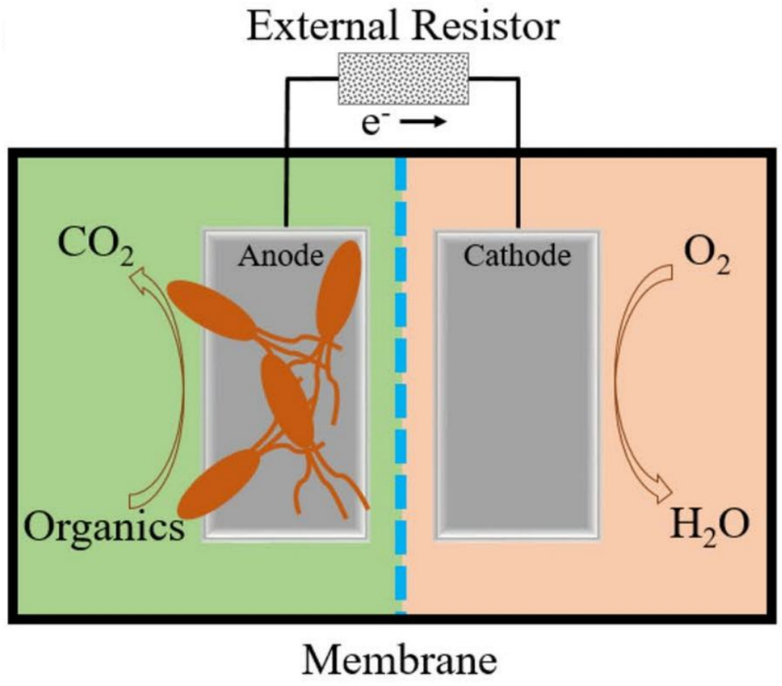
BES Technology

Definition

Innovative technologies that use microorganisms to treat wastewater, producing electricity and recovering valuable resources.

Core Principle

Microorganisms oxidize pollutants at the anode, generating electrons that power reduction reactions at the cathode.



Advantages and Drawbacks of BES Technology

Advantages

- ✚ Energy Generation: BES can produce electricity during pollutant degradation.
- ✚ Simultaneous Treatment and Recovery: Removes contaminants while recovering energy, nutrients.
- ✚ Eco-Friendly: Operates with minimal chemical inputs and low carbon emissions.
- ✚ Modular Design: Easily scalable and adaptable to different settings.

Drawbacks

- ➖ High Costs: Expensive electrode materials like platinum increase construction costs.
- ➖ Low Power Output: Current BES designs generate limited electricity, unsuitable for energy-intensive applications.
- ➖ Membrane Fouling: Ion exchange membranes clog or degrade, reducing efficiency.

BES Configurations:

Microbial Fuel Cells (MFCs):
- Function: Treat organic contaminants while generating power (up to 3.64 W/m²)
- Efficient for municipal and industrial wastewater.

Microbial Desalination Cells (MDC)
- Function: Desalinate water while treating pollutants.(up to 0.2-0.7 W/m².)
- Ideal for brackish and coastal applications.

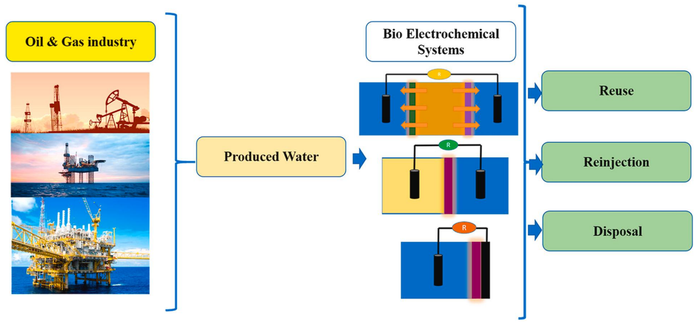
Microbial Recycling Cells (MRC)
- Function: Recovers nutrients like nitrogen and phosphorus.
- Ideal for rainwater reuse in agriculture and landscaping.

Application Steps

Primary Treatment: Primary Treatment: Organic breakdown via MFCs.

Desalination: Ion removal with MDCs.

Post-Treatment: Activated carbon filtration and UV disinfection for drinking standards.



Future Opportunities:

Hybrid Systems: Combining BES with traditional filtration for enhanced efficiency.

Cost Reduction: Exploring biochar electrodes to lower expenses.

Scalability: Developing modular designs for large-scale applications.

Nanotechnology Purification

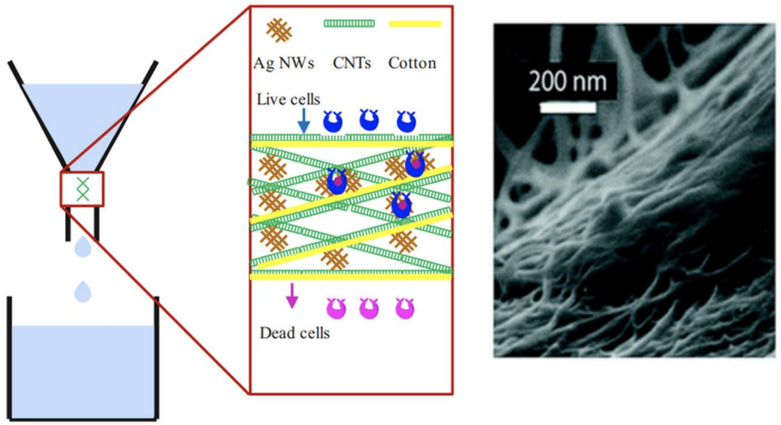


Fig. 8. Illustrates the cotton, silver nanowire, and CNT-based device's design, manufacturing method, and structure [136].

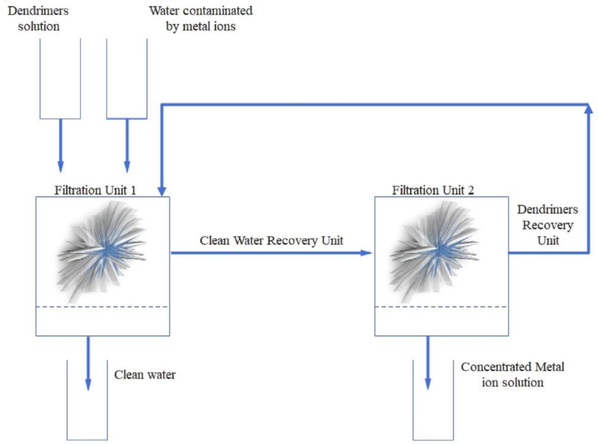


Fig. 4. Metal ion recovery from aqueous solutions using improved dendrimer filtration.

https://www.sciencedirect.com/science/article/pii/S2773207X23000271#sec2

Nanomaterials

There is a high variety of Nano materials that can be used to deal with different types of pollutants and microbes. The key ones we looked into were the use of nano-meshes to disinfect the water, and the use of Dendrimers to remove metal ions from the solution.

Nano-mesh

The nano-mesh (or nano-membrane) is a passive filter made of carbon nanotubes, cotton and silver wire, through which as water comes through, the mesh structure traps small particulates and sediments, while the silver particulates inside the mesh kill microbes within the water.

While this system works well, key issues exist around fouling of the mesh (aka. it gets clogged up with particulates) resulting in efficiency going down. However, these can be cleaned with back flushing.

Dendrimer Filtration

Dendrimer are a type of nanostructure which attract and bind metal ions purifying the water. However, they need to be removed from the water and then recovered to make the system closed loop. They also take very long to filter water, up to 48 hours.

Future Development

Nanotechnology holds significant promise for addressing global water challenges by enhancing purification methods, reducing costs, and overcoming limitations of traditional systems.

MIT Desalination System

Objective

Using direct-drive photovoltaic electrodialysis (PV-ED) system [Powered by the sun] using a simple (FCCC) strategy for renewable-powered brackish groundwater desalination.

This allows for desalination with minimal energy storage which helps lower cost, maintenance, and complexity when comparing with traditional desalination techniques.

Performance

Using PV energy, **94% utilisation** of water was achieved with **99% reduction** in energy storage requirements.

Daily water production reached **5.29m^3** with an energy consumption of **1.58 kWh/m^3** and a cost of **$6.37/m^3.** This makes it highly competitive against traditional desalination systems such as Reverse Osmosis (RO).

Applications and development

The system was tested for six months under real-world conditions in New Mexico, dealing with solar irradiance and groundwater salinity. The system maintained target salinities of less than **1,000 µS/cm**, meeting drinking water standards.

FCCC can be extended to other electrochemical applications, such as carbon capture, high-salinity water treatment, and fuel cell operations, or to remove large pollutants like arsenic in the water.

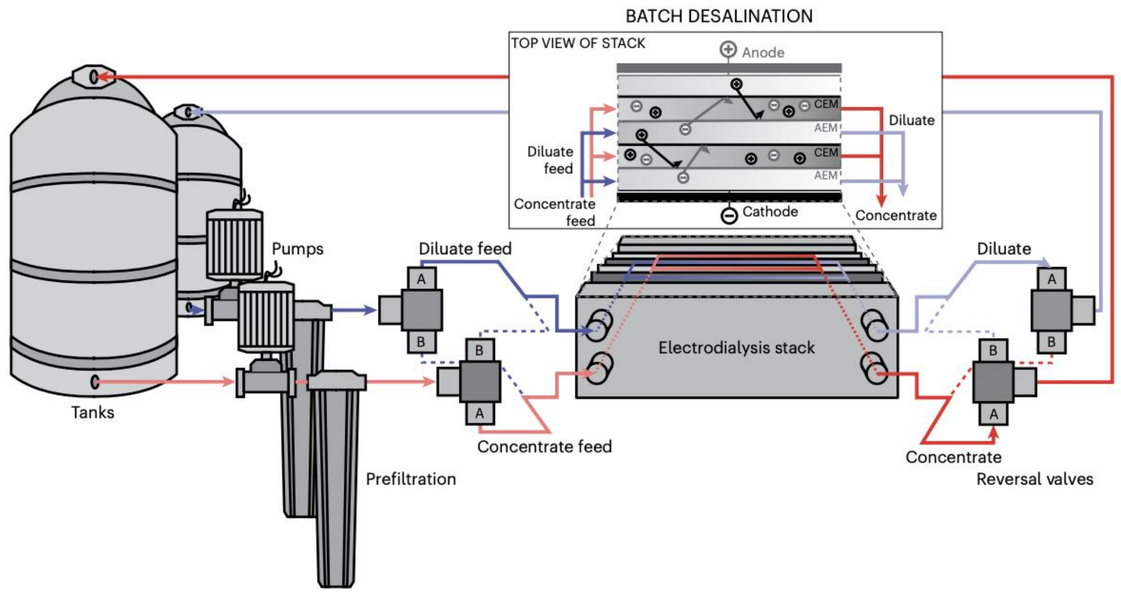


Fig. 1 | Conceptual diagram of a batch electrodialysis system. Where water is fed from batch tanks via parallel pumps into optional cartridge filters and then directed through a reversal valve network into the electrodialysis stack. In the electrodialysis stack, ions are transferred across selectively charged membranes: including pairs of a cation exchange membrane (CEM) and anion exchange membrane (AEM). The labels on the valve networks show potential hydraulic pathways, where in this depiction, channel configuration 'A' is engaged and 'B' is disengaged.

Other Enablers

Other | Water Reclamation | Water Transport | Water Purification

Plasma-Based Purification:

- https://roxia.com/articles/plasma-oxidizer-water-treatment-technology/
- The system contains electrode stacks that generate special types of electric discharges to create an ambient-temperature plasma field inside a reactor.
- The water to be treated is then allowed to pass through the plasma volume with high specific plasma-water contact surface area. Maximising the contact surface area allows for abundant OH-radicals to be formed on the interface, contributing to effective oxidation of dissolved pollutants.
- VERY COMPLICATED - READ DOCUMENT

Forward Osmosis (FO) Systems:

- https://aquaporin.com/what-is-forward-osmosis/
- A water separation process which uses a semipermeable membrane and the natural energy of osmotic pressure to separate water from dissolved solutes. The osmotic pressure is used to transport water through the membrane while retaining all the dissolved solutes on the other side.
- In both product concentration and wastewater concentration, forward osmosis moves water from the feed solution into the draw solution. This water can then be recovered to produce clean water using a draw solution regeneration step such as reverse osmosis.

Autonomous Vehicles and Robotics:

Autonomous drone and robotic systems in logistics and delivery could inspire automated, unmanned water maintenance tools. Imagine autonomous water inspection drones or underwater robots that maintain pipes or reservoirs without human intervention.

Bioluminescent Water Quality Indicators

In a future city, all water pipes would contain genetically engineered bioluminescent organisms that glow to indicate water quality. If contaminants are detected, the glow changes color instantly, so water quality issues are visible in real time. This concept would extend even to rivers and lakes, where bioluminescent organisms would create glowing riverscapes to indicate areas safe for drinking or swimming.

Closed-Loop or "Zero Liquid Discharge" Systems

Food and beverage companies are increasingly In water-scarce areas, some plants operate closed-loop systems, where all water is treated and reused on-site without any discharge. Zero liquid discharge (ZLD) systems use advanced treatment to recover and recycle every drop. These systems would allow facilities to use water repeatedly with minimal external input, creating nearly self-sustaining water cycles that also prevent polluted water from reaching rivers and oceans.

Underground River and Aquifer Pumping Networks:

- Using Mexico city's extensive aquifer network, fresh water could be moved around below the city without needing to use pipes.
- We could then introduce a purification method within it to allow this water to be used.
- Look at qanāts in Iran.

Balloon-Based Water Transport (Water Bladders):

- https://www.ijream.org/papers/ICRTET0037.pdf
- Large, flexible bladders (or balloons) are filled with freshwater and towed across water by tugboats. These bladders float on the surface and can carry massive volumes.
- Cost-effective, minimizes evaporation, and can be deployed quickly during droughts or emergencies.

Wireless Water:

- Transport of water along long distances wirelessly
- Very scientific product that could be used in the future.
- Tried and tested with the bluetooth water hose:
  - https://www.youtube.com/shorts/yo_g6nIEO1U

Material: Self-healing Concrete

- Can repair internal cracks by themselves, without human intervention
- Autogenous healing: originated naturally from cementitious material
- Autonomous healing: relies on embedded unconventional engineered additions
- Autonomous uses electrodeposition tech and embed shape memory alloy (SMA)

(source)

Robot: Swarm Robotics

- groups of simple homogeneous robots, coordinated in a distributed and decentralised way
- motivated by the striking behaviour of social insects and other swarms in nature, which are able to achieve complex tasks using only local communication and sensing capabilities

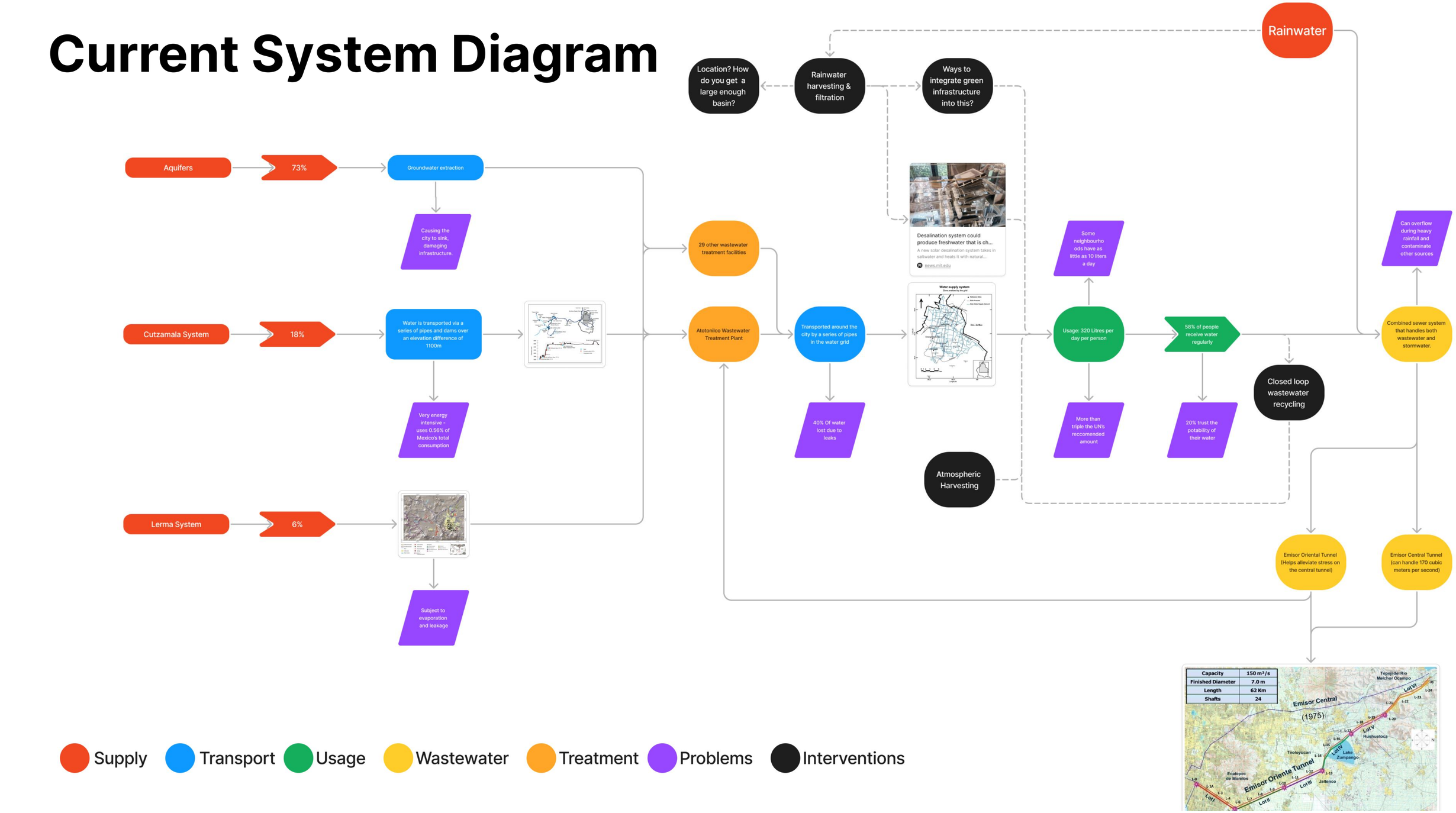
(source)

Metal-Organic Frameworks (MOFs)

1. Portion of the microporous MOF initially absorbs water using unsaturated metals or polarity sites on ligands,
2. Initially adsorbed water molecules continuously attract water to form water clusters or networks through hydrogen bonds

(source)

Current System Diagram



20 Initial Ideas

Local / Residential Level Solutions

Water purification

Pipe System

Water generation

transforms low-lying roads into flood-resilient, groundwater-replenishing systems by purifying and monitoring rainwater through multi-layer filtration, bio-sensors, and bio-coated pipelines.

Robot balls that float around the pipes, purifying water and sealing up cracks

Nanomesh filters in peope's homes to make water more drinkable

autonomous drone system to deliver desalinated seawater directly to communities → could be an underground tunnel system

Inner pipe lining with bio filtration material

Decentralised, emergent system with microbial fuel cells (detects leakage / damage in pipe and manage the amount of flow locally)

Zero discharge closed loop system for the homes, like they have on the ISS

New, malleable & self-healing pipeline system equipped w/ sensors for data collection & leak/ illegal pumping detection

Porous material with filtration, bringing water back to ground (tech to balance it out evenly)

Using the underwater aquifers to act as pipes or underwater rivers, where water can then be purified and pumped into homes from it

Utilising trucks or trains to transport water

Decentralised atmospheric water collection system

Underground Intelligent Water Retention Tunnel System

Wireless water hose

Rooftop rainwater collection and pumping into aquifer

Water purification by desalination process

Pipe system redirection: bio-sensing of contaminant → stop the flow in main pipe and activate local system

Wastewater Resource Recovery and Microbial Fuel Cell System

Repair Robots (tube-shaped) Identifies and repairs the leakage

Water harvesting from the atmosphere

Design Objectives

A

Durability

Withstand the environmental conditions, enabling persistent use with minimal maintenance

B

Affordability

Total cost of the system reach for low to medium income households and/or small communities in Mexico City

C

Scalability

Capable of addressing the structural demand and configurations

D

Ease of Use

Minimal user intervention in operation

E

Ease of Installation

Requires minimal set up and installation process

F

Water Quality

Processed water meets the standard for potable water

G

Decentralisation

Operate independently, resilient to disruption in central system

H

Size

Suitable to spaces within household units

I

Monitoring & Control

Monitors the data on the system → internal feedback for control? or communicate?

Concept Evaluation

		A	B	C	D	E	F	G	H	I	Score	Weight	Concept 1: Porous pavement	Concept 2: Rooftop Moss Catcher	Concept 3: Desalination
Durability	A		B	A	D	E	F	A	A	A	4	11.11%	3	2	1
Affordability	B			B	D	E	F	G	B	B	4	11.11%	1	3	3
Scalability	C				D	E	F	G	H	C	1	2.78%	2	3	3
Ease of Use	D					D	F	D	D	D	7	19.44%	3	2	2
Ease of Installation	E						F	G	H	E	4	11.11%	1	2	2
Water Quality	F							F	F	F	8	22.22%	3	3	3
Decentralisation	G								G	G	5	13.89%	1	2	3
Size	H									I	2	5.56%	3	2	2
Monitoring & Control	I										1	2.78%	3	1	1
												Total	2.25	**2.47**	2.36

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DeepAI: HD/Quality/Fantasy setting. prompt; dry Mexico city ally, infrastructure, people

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Technical Research

Energy Generation: BES can produce electricity during pollutant degradation, providing a sustainable energy source.

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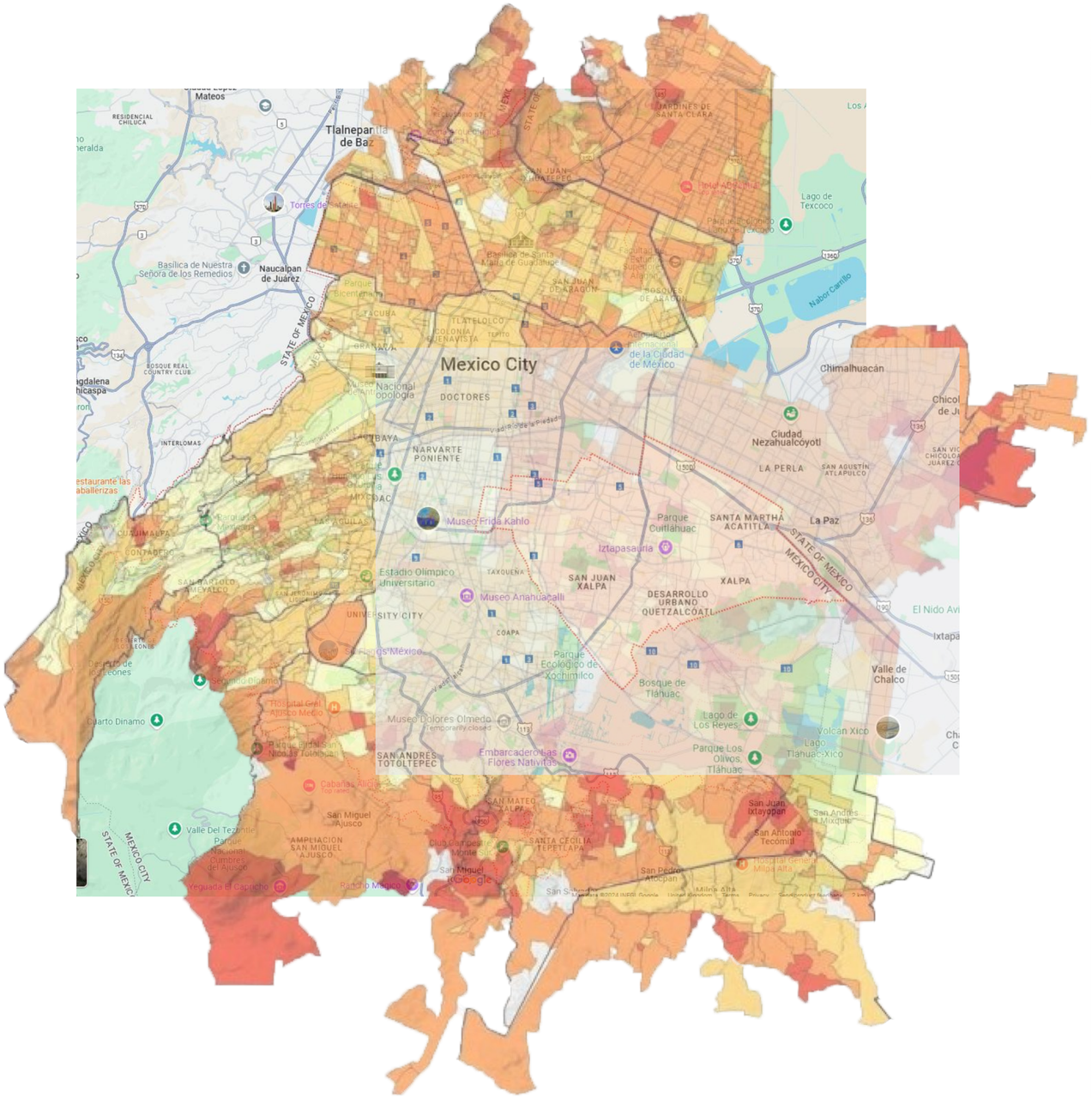
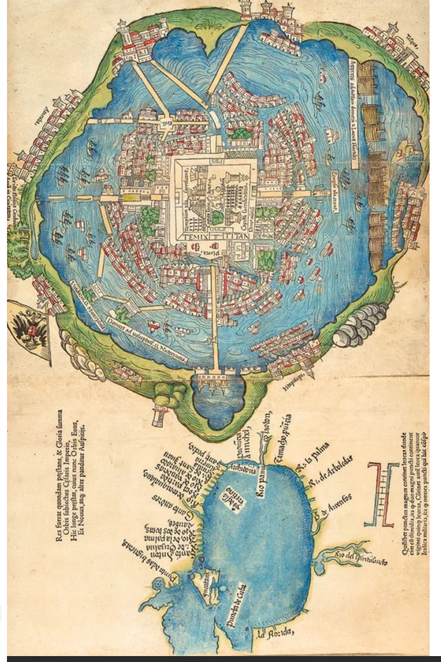
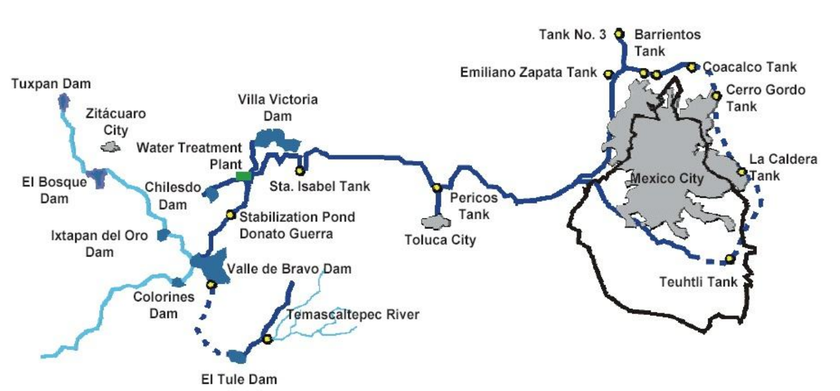
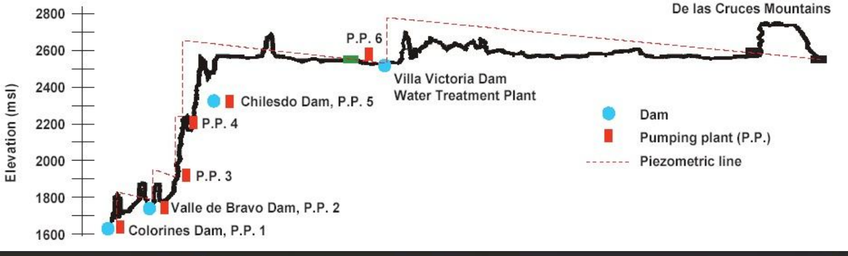

Mexico City
Tlalnepantla de Baz
Naucalpan de Juárez
Chimalhuacán
Ciudad Nezahualcóyotl
La Paz
Iztapalapa
Valle de Chalco
Museo Frida Kahlo
Museo Anahuacalli
Parque Ecológico de Xochimilco
Bosque de Tláhuac
Lago de Texcoco
STATE OF MEXICO
MEXICO CITY




Tuxpan Dam
Zitácuaro City
El Bosque Dam
Villa Victoria Dam
Water Treatment Plant
Chilesdo Dam
Sta. Isabel Tank
Stabilization Pond Donato Guerra
Ixtapan del Oro Dam
Valle de Bravo Dam
Colorines Dam
Temascaltepec River
El Tule Dam
Pericos Tank
Toluca City
Tank No. 3
Barrientos Tank
Emiliano Zapata Tank
Coacalco Tank
Cerro Gordo Tank
La Caldera Tank
Mexico City
Teuhtli Tank


Elevation (msl)
2800
2600
2400
2200
2000
1800
1600
De las Cruces Mountains
P.P. 6
Villa Victoria Dam
Water Treatment Plant
Chilesdo Dam, P.P. 5
P.P. 4
P.P. 3
Valle de Bravo Dam, P.P. 2
Colorines Dam, P.P. 1
Dam
Pumping plant (P.P.)
Piezometric line

PLANO DE TENOCHTITLAN
ATRIBUIDO A
HERNAN CORTES
(circa 1524)
2 700 000
30 000
111.10
Superficie total
(metros cuadrados)
Población
(núm. de habitantes)
Densidad
(habitantes por hectárea)
PLANO ACTUAL
DE LA
CIUDAD DE MEXICO
(cifras calculadas
para junio de 1964)
295 000 000
5 890 500
195.02
escala gráfica